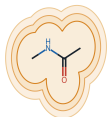


a SOLVATION RESPONSE

training information · generated once

Calculated solvation

COSMOtherm · SMD(water)



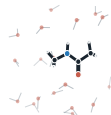
Polarity / H-bonding

Abraham E · S · A · B · L



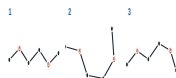
Water response

OpenFF explicit · GBn2 implicit



Conformational response

ConfSolv H₂O · six summaries



STAGE 1 learn structure → response



six frozen response surrogates

source calculations do not run again

b REUSABLE RESPONSE LAYER

15 interpretable scalar coordinates

15

named scalars

CALCULATED SOLVATION

- 1 COSMOtherm water response
- 9 SMD(water) response

POLARITY / H-BONDING

- 2 Abraham E · excess molar refraction
- 3 Abraham S · dipolarity / polarizability
- 4 Abraham A · H-bond acidity
- 5 Abraham B · H-bond basicity
- 6 Abraham L · hexadecane-air partition

EXPLICIT / IMPLICIT WATER

- 7 corrected OpenFF explicit-water response
- 8 corrected GBn2 implicit-solvent response

CONFORMATIONAL RESPONSE

- 10 gas conformer correction
- 11 solution conformer correction
- 12 hydration conformer correction
- 13 conformer solvation-energy spread
- 14 mean conformer solvent response
- 15 conformer response spread

STAGE 2 experimentally supervised endpoint

STRUCTURE

2,265 features

RESPONSE

15 coordinates

EXPERIMENT

Δ G_{hyd} labels

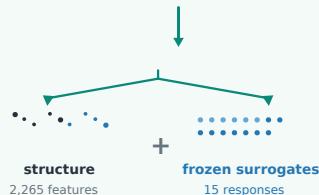
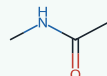
ExtraTrees endpoint ensemble

3 × 360 trees · frozen after evaluation

c STRUCTURE-ONLY

deployment · SMILES input

NEW MOLECULE



structure 2,265 features + frozen surrogates 15 responses



SolvAI

frozen endpoint ensemble

ΔG_{hyd}
kcal mol⁻¹

SMILES only · no MD · no PIMD · no probe

ARROW/PIMD8 ≈ 0.205 kcal mol⁻¹

accuracy reference only · not a retained teacher